

Something for nothing

Georgia Tech researchers have found a way to cheaply tap ambient electromagnetic energy from the air

Bionic guiding light

Oxford University scientists have developed bionic glasses to aid the visually challenged, using minuscule cameras to alert wearers about the things around them

Crysis 2 DX 11 Ultra Upgrade and Sandbox Editor

Crytek has been busy this month, giving Crysis 2 players a DirectX 11 Ultra Upgrade, allowing the game to use a powerful graphics card's full potential. It also released the CryEngine 3 Sandbox Editor. Crysis 2 players interested in the DX11 upgrade will first have to patch the game to version 1.9 (136MB), before installing the DirectX 11 Ultra Upgrade (545MB), and Hi-Res Texture Pack (1.65GB). The texture pack is meant for systems with a 64-bit operating system and a GPU with more than 768MB of video memory. Patch v1.9 should update automatically

once you start the game while connected to the internet.

New graphics features introduced for CryEngine 3 with the DirectX 11 Ultra Upgrade include Hardware Tessellation and Displacement Mapping, Parallax Occlusion Mapping, Hardware-based Occlusion Culling, HDR motion blur and post-processing, Improved Tone Mapping, Improved Water Rendering, Interactive Water, Realistic Shadows with Variable Penumbra, Particle Motion Blur and Shadows, Custom Shape Based Bokeh Depth of Field, and Real-time Screen Space Directional Occlusion (contact shadows) on top of the already



present SSAO. A new and more complete advanced graphic menu has been added in-game. Also promised, is improved multi-GPU support, which makes us worry about single-card owners like ourselves.

The new CryEngine 3 Sandbox Editor, allowing them to create their own maps, mods, and content for the game. Giving a fair amount of control,

Crytek will be providing additional tools with the package, including plugins for Maya and 3ds Max, Poly Bump, FMOD Designer, and more. For beginners, sample content has also been provided to help build and edit. The crymod.com mod community has also been re-opened, allowing fellow Crysis 2 modders to share and discuss their work. ■

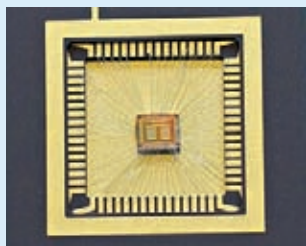
IBM develops multi-bit phase change memory

IBM Research – Zurich has announced a new type of inexpensive and reliable universal non-volatile memory, called phase change memory (PCM). Shattering records, it supposedly reads and writes 100 times faster than conventional flash-based memory with a latency of 10 microseconds, and is also several of orders of magnitude more reliable, with 10 million estimated write-cycles.

Surprisingly, the technology behind phase change memory is not expensive, and can be commercially used for a variety of devices requiring storage and memory, from servers to tablets. It's not going to be hitting markets very soon, with IBM stating the technology would enable "a paradigm shift for enterprise IT and storage

systems, including cloud computing by 2016."

IBM's phase-change memory is built using cells created from a spe-



cial alloy, which can be electrically controlled into different physical states – crystalline with low resistance properties, amorphous with high resistance properties. Each alloy cell is capable of storing four data bits using different resistance levels, making it the first multi-bit phase-change memory with reliable retention – only single bit PCM was capable of this earlier. ■

Zapak launches subscription based gaming services

Popular gaming website, Zapak.com, has launched a subscription-based gaming service – Zapak Plus, enabling users can download full games on their personal computers and play them via a client, by subscribing to monthly or quarterly plans. Initially, the gaming company is offering around 120 titles from different genres such as arcade, sports, action and others.

Charges for now are fixed at ₹99 for a month of playtime, or ₹247 for three months. Rohit Sharma, chief executive officer, Reliance Entertainment (Digital Business), says, "Zapak Plus offers an opportunity to millions of gamers

to experience some of the biggest titles in PC gaming at less than Re 1 per month, in the comfort of their homes. Zapak is now a destination not only for flash games but for PC games, too. Gamers, who hitherto had access only to highly-priced



gaming DVDs or poor quality of pirated game DVDs, can subscribe to Zapak Plus and enjoy good quality full-version PC games at affordable pricing." ■